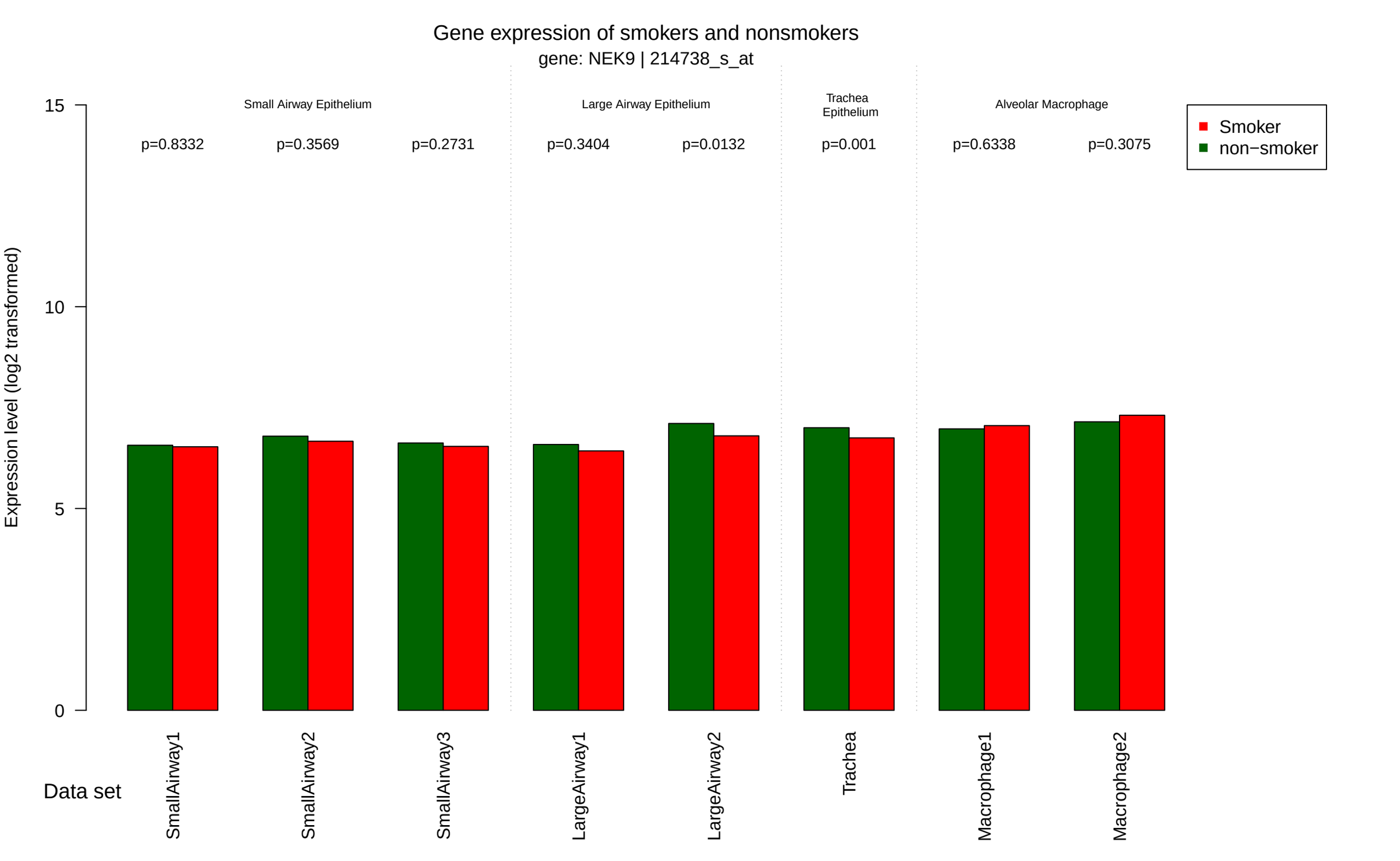
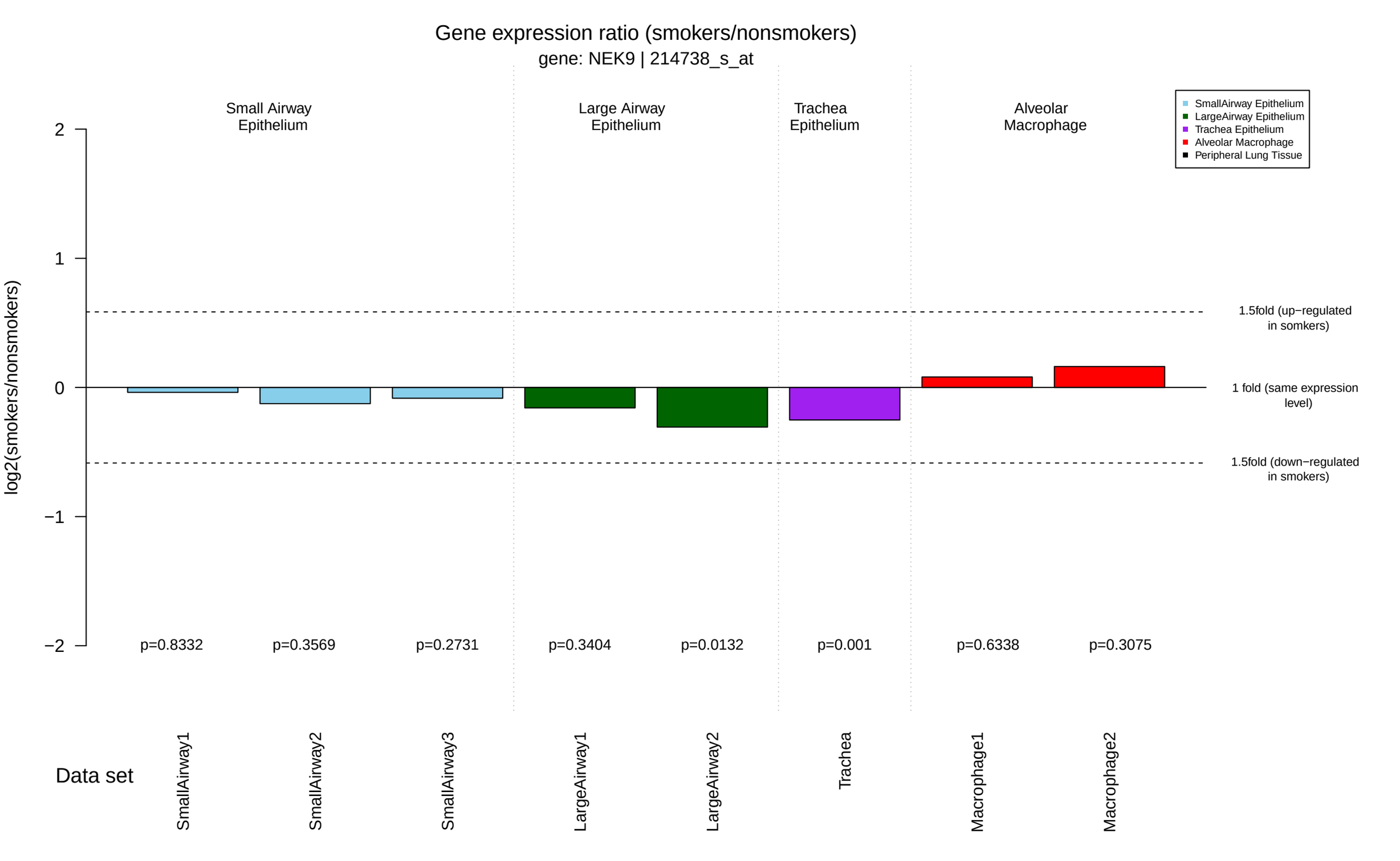
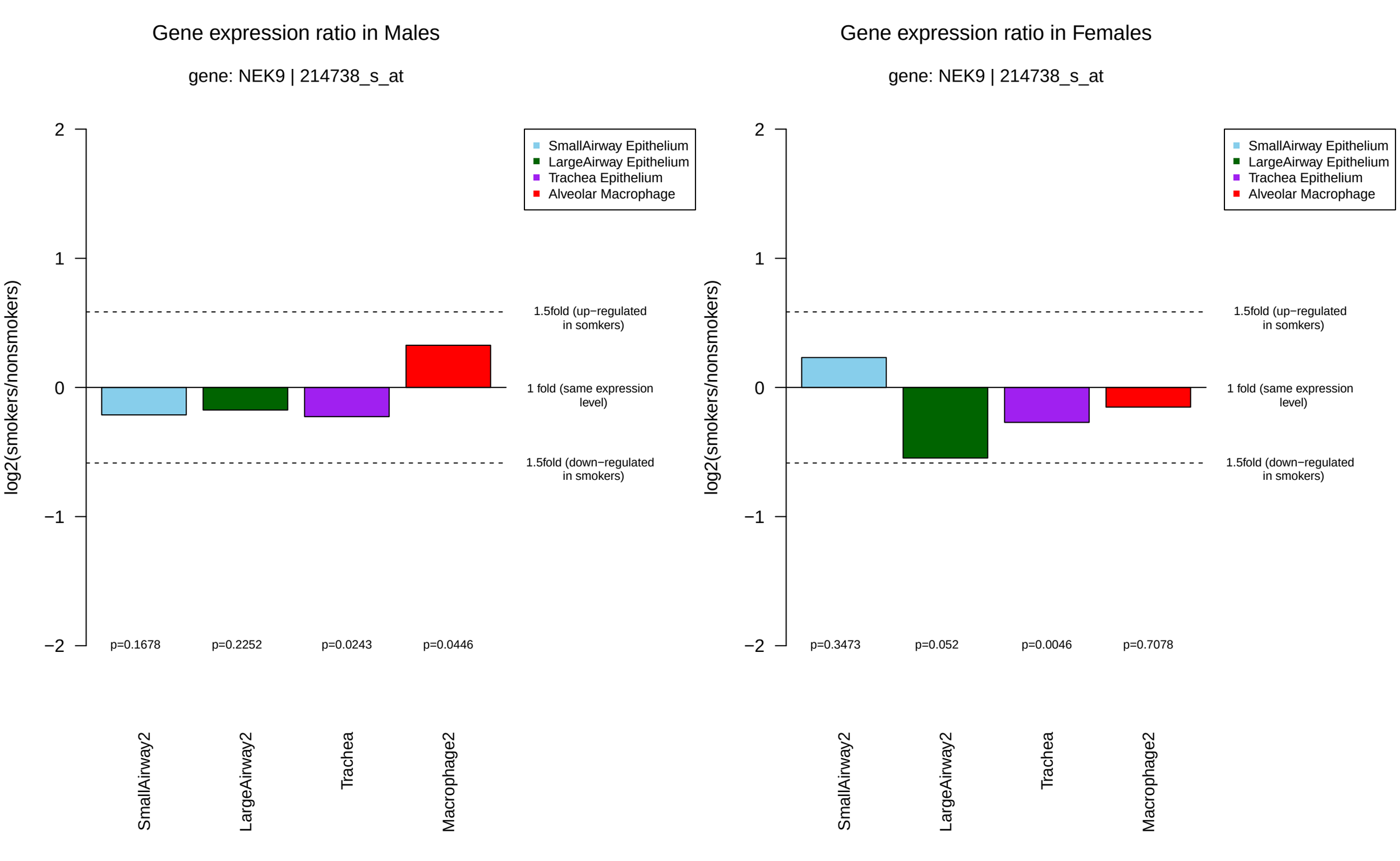
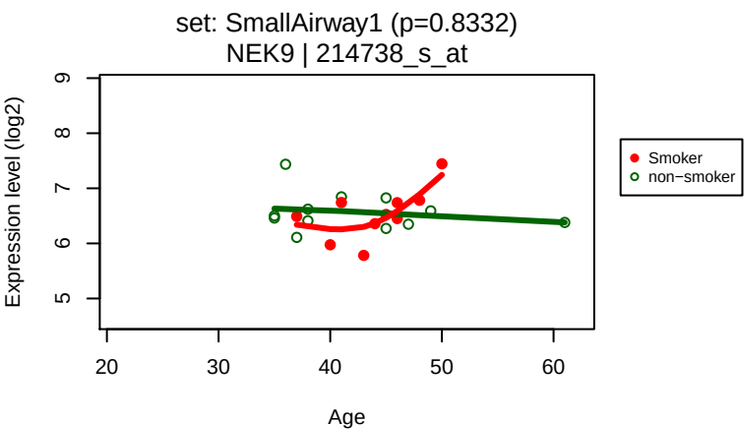
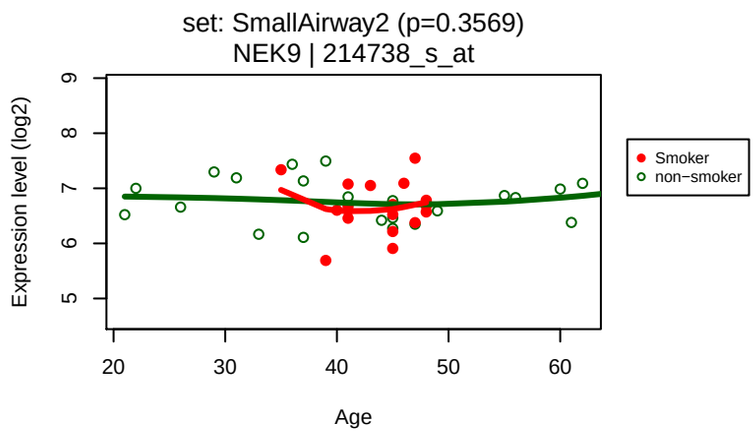
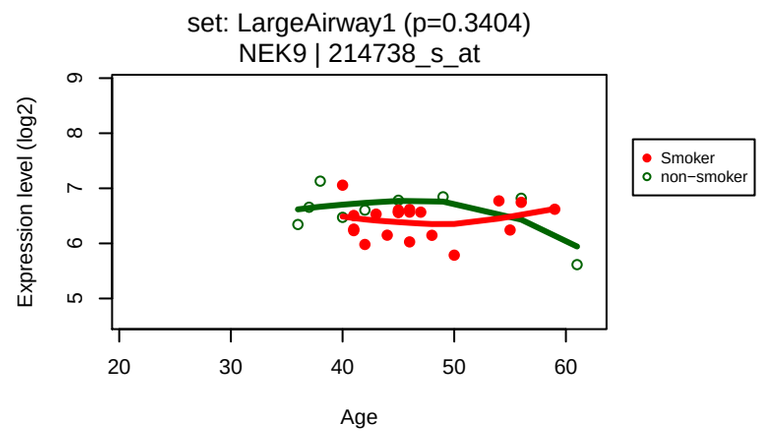

Gene expression of smokers and nonsmokers
gene: NEK9 | 214738_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8332
p=0.3569
p=0.2731
p=0.3404
p=0.0132
p=0.001
p=0.6338
p=0.3075
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

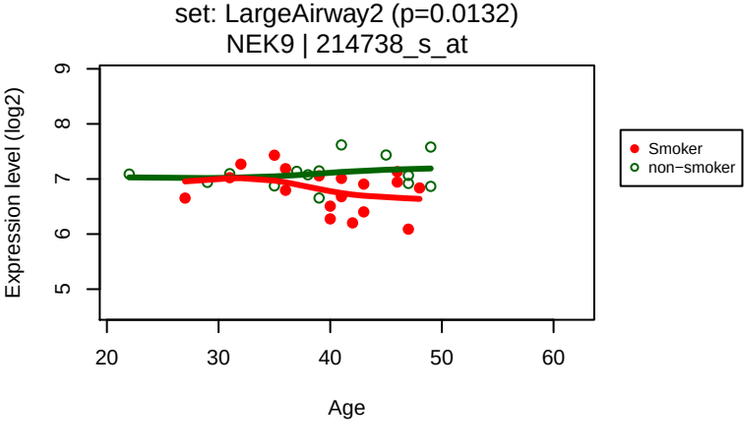

Gene expression ratio (smokers/nonsmokers)
gene: NEK9 | 214738_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8332
p=0.3569
p=0.2731
p=0.3404
p=0.0132
p=0.001
p=0.6338
p=0.3075
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

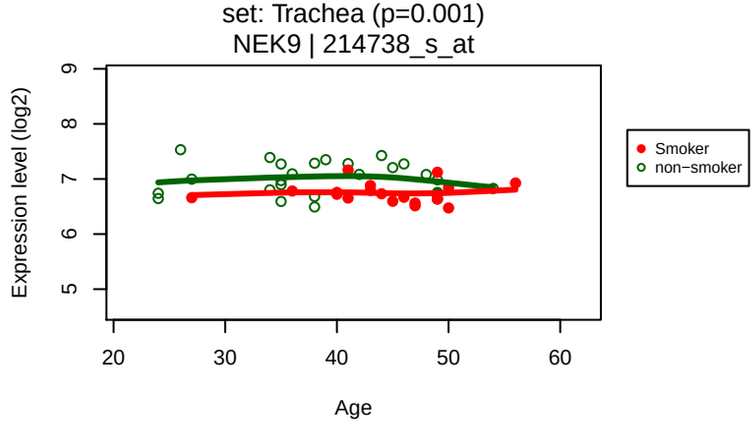

Gene expression ratio in Males
gene: NEK9 | 214738_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1678
p=0.2252
p=0.0243
p=0.0446
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NEK9 | 214738_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3473
p=0.052
p=0.0046
p=0.7078
SmallAirway2
LargeAirway2
Trachea
Macrophage2

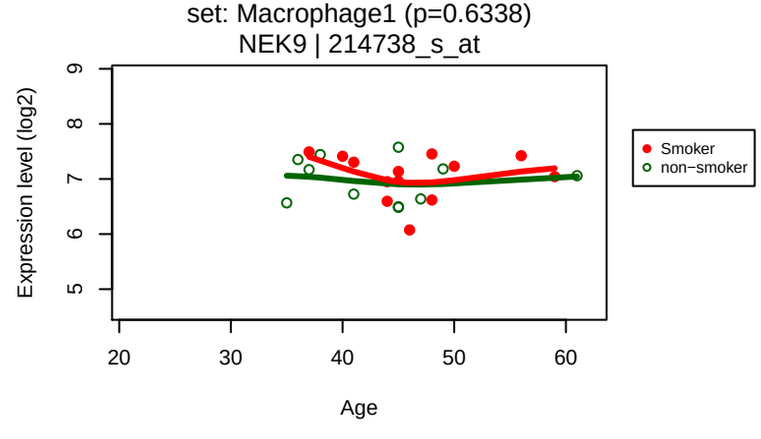

set: SmallAirway1 (p=0.8332)
NEK9 | 214738_s_at
Expression level (log2)
Age
Smoker
non−smoker

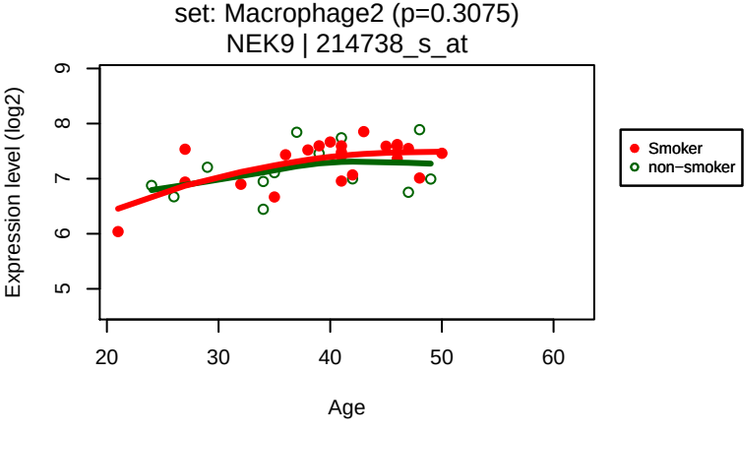

set: SmallAirway2 (p=0.3569)
NEK9 | 214738_s_at
Expression level (log2)
Age
Smoker
non−smoker

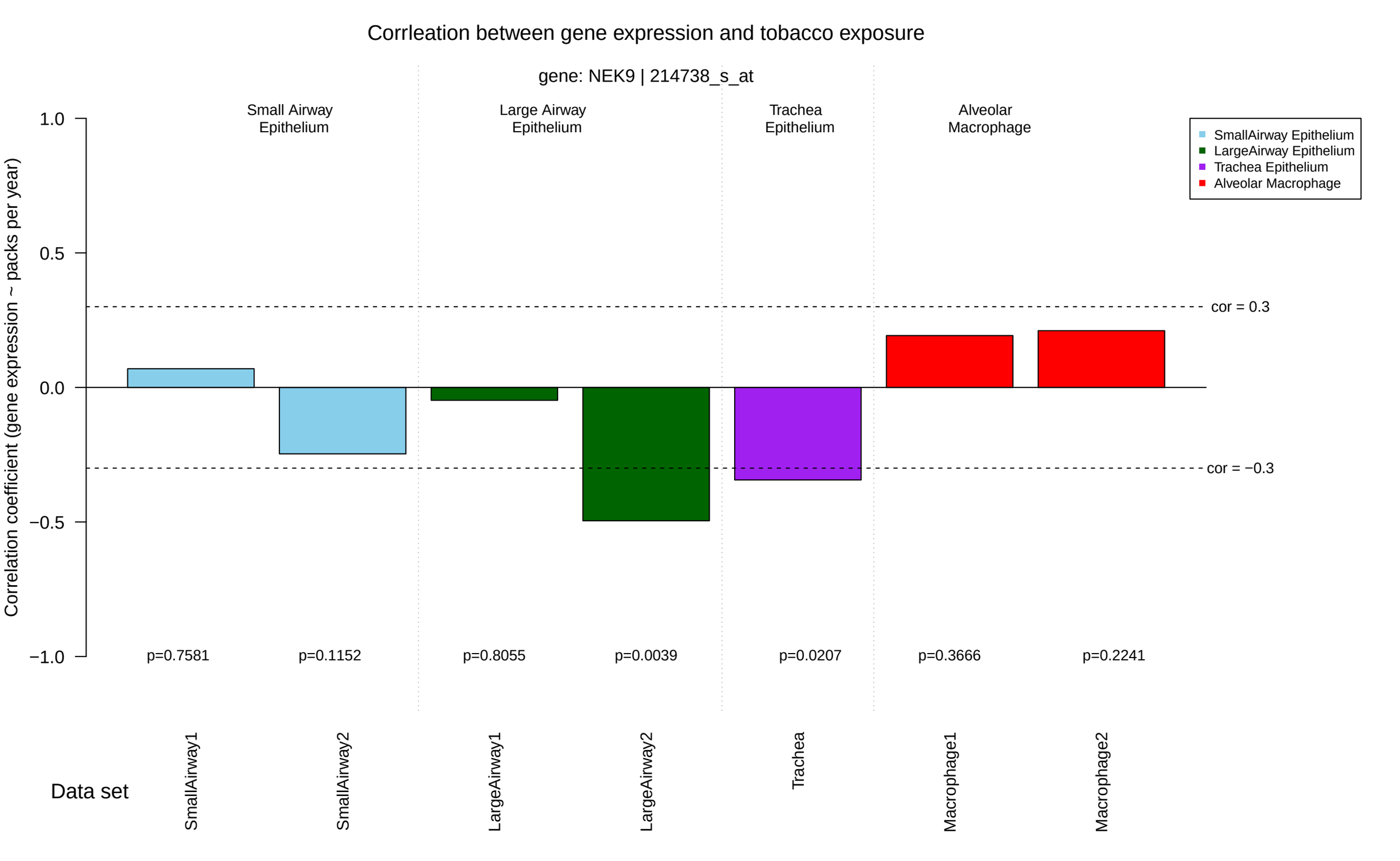

set: LargeAirway1 (p=0.3404)
NEK9 | 214738_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0132)
NEK9 | 214738_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.001)
NEK9 | 214738_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6338)
NEK9 | 214738_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3075)
NEK9 | 214738_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NEK9 | 214738_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7581
p=0.1152
p=0.8055
p=0.0039
p=0.0207
p=0.3666
p=0.2241
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2